

Daily Number Theory Drill #6

Sheet by Agent Antigravity

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: The Euclidean Step · Polynomial Division · GCD Logic.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Prime factorisation, divisibility rules, and modular arithmetic basics.

- Find the largest prime factor of 2025.
- What is the last digit of 3^{2026} ?
- Find the greatest common divisor of 84 and 120.
- What is the remainder when 10^{10} is divided by 9?
- Find the missing digit d such that the five-digit number $47d53$ is divisible by 9.
- How many positive integer factors does 100 have?
- Which is larger: 2^{300} or 3^{200} ?
- Evaluate $5^{2024} \pmod{4}$.
- Find the smallest positive integer n such that $12n$ is a perfect square.
- What is the sum of the prime factors of 210?

Section B Manipulation Drills

(≤50 s each)

Linear congruences, powers of primes, and prime structure.

- Find the number of trailing zeroes of $20!$.
- Solve for x in $3x \equiv 4 \pmod{7}$, with $0 < x < 7$.
- If p and q are prime numbers such that $p^2 - q^2 = 24$, find the value of $p + q$.
- Find the last two digits of 5^{2025} .

5. A number N leaves a remainder of 2 when divided by 3, and 3 when divided by 5. What is the smallest positive integer value of N ?
6. How many perfect squares divide 3600?
7. Find the highest power of 3 dividing $27^5 \times 9^4$.
8. What is the smallest positive integer with exactly 6 positive divisors?
9. If x and y are positive integers such that $xy = 144$ and $\gcd(x, y) = 3$, find the maximum possible value of $x + y$.
10. Calculate the remainder when $1! + 2! + 3! + \cdots + 10!$ is divided by 5.

Section C Substitution & Structure**(≤ 90 s each)**

Synoptic links: use algebra and combinatorics alongside Diophantine logic.

1. Find the number of pairs of positive integers (x, y) such that $xy + x + y = 23$. (after SMC 2018 Q16)
2. Find the sum of all prime numbers p such that $p^2 + 2$ is also prime. (after SMC 2016 Q20)
3. How many 3-digit palindromes are divisible by 11? (after SMC 2015 Q19)
4. The product of three consecutive positive integers is equal to 336. Find the sum of these integers. (after SMC 2012 Q14)
5. Let N be a 4-digit number. When its digits are reversed, the new number is $N + 1089$. If the sum of the digits of N is 12, what is the maximum possible value of N ? (after SMC 2017 Q21)
6. Find the sum of all positive integers x such that $\frac{x+11}{x-3}$ is an integer. (after SMC 2019 Q18)
7. Find the maximum possible value of $\gcd(7n + 15, 3n + 2)$, where n is a positive integer. (after SMC 2013 Q21)
8. Find the sum of all integers n such that $\frac{n^2+3n+5}{n-1}$ is an integer. (after BMO1 2007 Q1)

Section D Challenge**(BMO1 difficulty)**

Insight over computation. Uncover the structure.

1. Find all three-digit numbers abc such that $abc = a! + b! + c!$. (after SMC 2004 Q21)
2. Let $d_n = \gcd(n^2 + 3, n + 1)$. Find the sum of all possible values of d_n as n ranges over all positive integers. (after SMC 2010 Q24)
3. Find the last three digits of 2025^{2025} . (after SMC 2008 Q24)

4. For how many positive integers $n \leq 100$ is n^n a perfect square? (*after SMC 2001 Q22*)
5. Find the maximum value of $x + y + z$ given that x, y, z are positive integers satisfying $x \leq y \leq z$ and $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$. (*after BMO1 2011 Q1*)

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths